

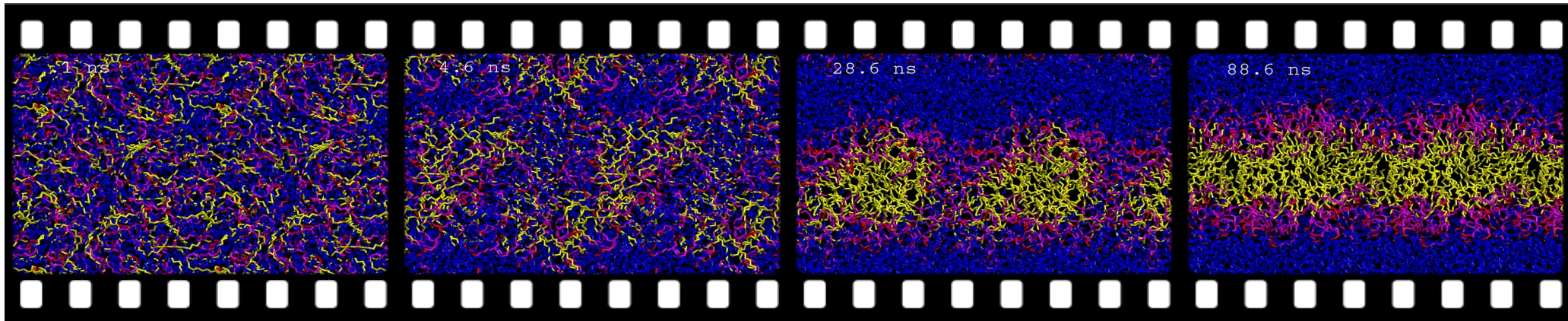


Science ouverte, données et IA : automatiser l'annotation scientifique

Pierre Poulain

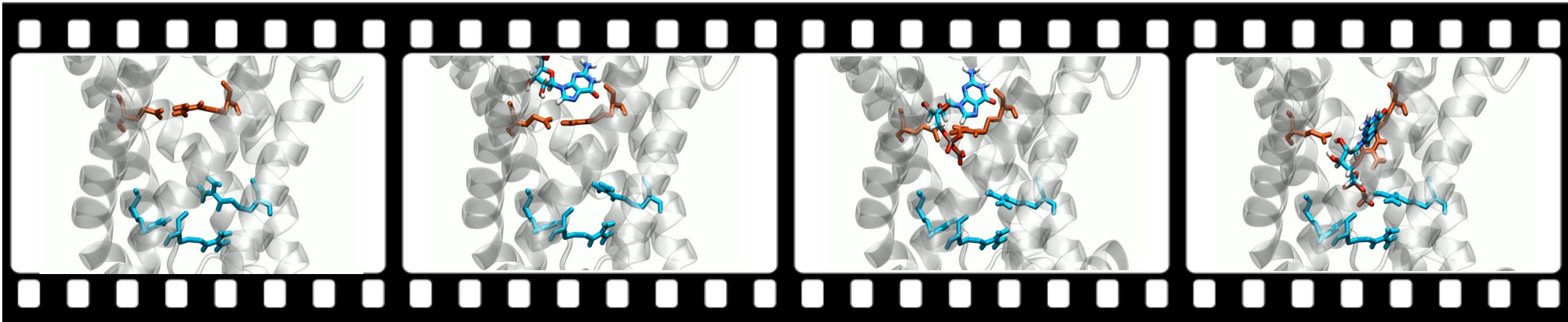
Laboratoire de Biochimie Théorique
UMR 8266 CNRS & Université Paris Cité

What is molecular dynamics (MD)?



water + detergent

Senac et al, Langmuir, 2017. Movie by Patrick Fuchs.



Gagelin et al, Nature communications, 2023.

Supercomputers → high cost



Source : Photothèque CNRS/Cyril Fréssillon (droits réservés)

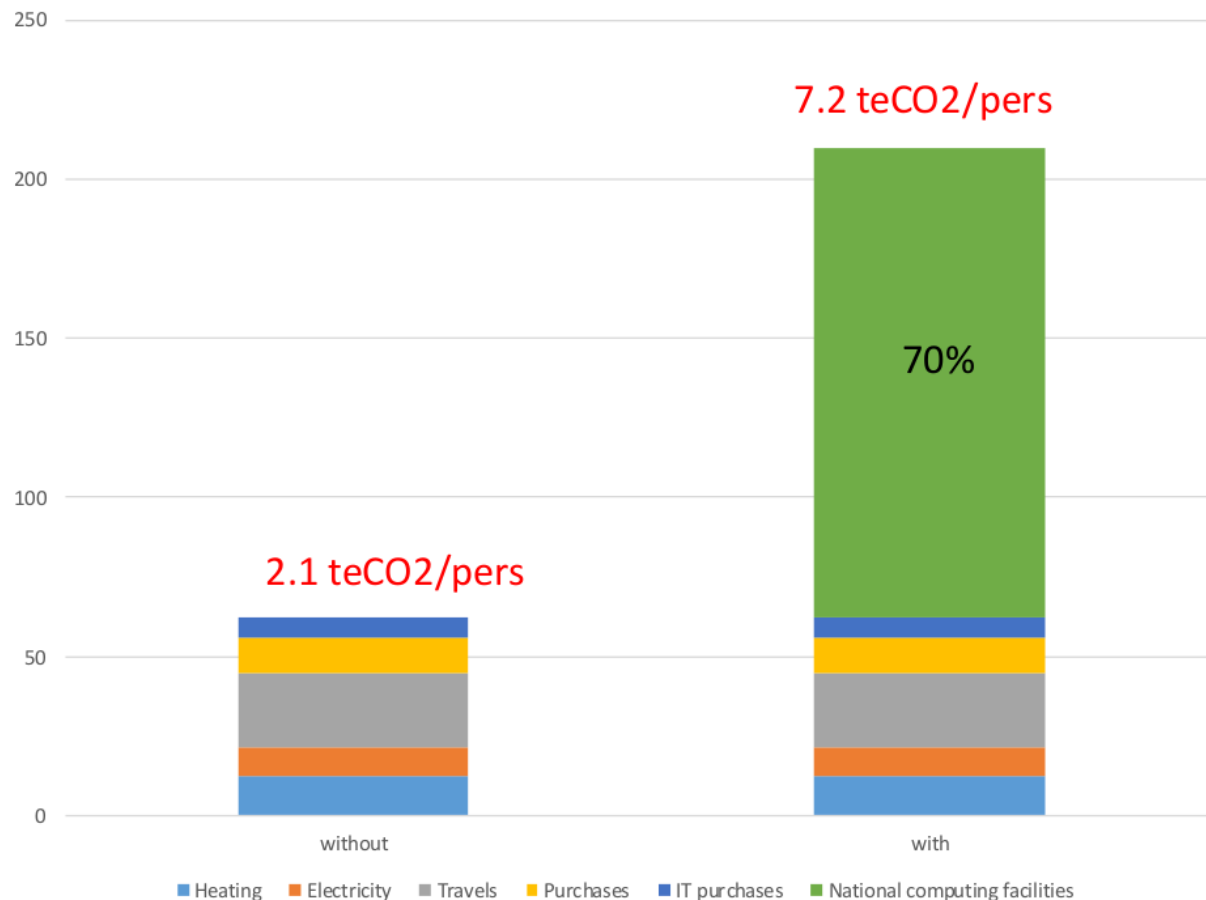


In 2025:

- CT 7 (simulation in biology)
- 132 Mh CPU
- 8 Mh GPU
- **Total cost ~8 M€**

Patrick Fuchs, 2026.

Supercomputers → high environmental cost



Welcome to the Dark Matter of MD

Data that is **technically accessible**,
but neither **indexed**, **curated**,
or easily **searchable**.



RESEARCH ARTICLE

MDverse, shedding light on the dark matter of molecular dynamics simulations

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Tiemann et al. eLife 2023;12:RP90061. DOI: <https://doi.org/10.7554/eLife.90061> 1 of 22

Abstract The rise of open science and the absence of a global dedicated data repository for molecular dynamics (MD) simulations has led to the accumulation of MD files in generalist data repositories, constituting the *dark matter of MD* — data that is technically accessible, but neither indexed, curated, or easily searchable. Leveraging an original search strategy, we found and indexed about 250,000 files and 2000 datasets from Zenodo, Figshare and Open Science Framework. With a focus on files produced by the Gromacs MD software, we illustrate the potential offered by the mining of publicly available MD data. We identified systems with specific molecular composition and were able to characterize essential parameters of MD simulation such as temperature and simulation length, and could identify model resolution, such as all-atom and coarse-grain. Based on this analysis, we inferred metadata to propose a search engine prototype to explore the MD data. To continue in this direction, we call on the community to pursue the effort of sharing MD data, and to report and standardize metadata to reuse this valuable matter.

eLife assessment The study presents a **valuable** tool for searching molecular dynamics simulation data, making such datasets accessible for open science. The authors provide **convincing** evidence that it is possible to identify noteworthy molecular dynamics simulation datasets and that their analysis can produce information of value to the community.

Introduction The volume of data available in biology has increased tremendously (Marx, 2013; Stephens et al., 2015), through the emergence of high-throughput experimental technologies, often referred to as -omics, and the development of efficient computational techniques, associated with high-performance computing resources. The Open Access (OA) movement to make research results free and available to anyone (including e.g. the Budapest Open Access Initiative and the Berlin declaration on Open Access to Knowledge) has led to an explosive growth of research data made available by scientists (Wilson

A data catalogue for MD simulation data (and more)

MDverse

MDverse Search

I'm Feeling Lucky

A data catalogue for MD simulation data (and more)

MDverse

MDverse Search

I'm Feeling Lucky

WHY? Enable, facilitate & foster the reuse of MD data

MD data availability

Data repository	First dataset	# datasets	# files	Size (GB)
Zenodo	2014	2 369	504 891	26 686
Figshare	2012	1 373	222 703	1 177
NOMAD	2021	16 119	555 990	2 189
MDposit CINECA (MDDDB)	2025	310	251 581	52
MDposit INRIA (MDDDB)		719	186 293	1 022
MDposit MMB (MDDDB)		4 050	411 854	15 177
GPCRmd	2017	830	2 414	624
ATLAS	2023	1 938	69 768	10 944
Total		27 708	2 205 494	57 809

Reuse MD data → **Extract & normalize metadata**

Where to find metadata?

MD simulations of P-glycoprotein started from three different crystal structures: 3G5U, 4M1M and 4KSB

> <https://doi.org/10.6084/m9.figshare.4806544>



Cite

Download all (2.55 GB)

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Version 4 ▾

Dataset posted on 2018-01-10, 07:17 authored by [Karmen Condic-Jurkic](#)

Molecular simulations of mouse P-glycoprotein started from three different crystal structures with PDB IDs corresponding to 3G5U, 4KSB and 4M1M. The protein was transferred to POPC/cholesterol bilayer, followed by energy minimization and 10 ns equilibration period. All the simulations were performed in Gromacs 3.3.3 in conjunction with GROMOS54a7 force field at 300 K and pressure of 1 bar. Each system was run in triplicates for 200 ns.

USAGE METRICS

2341

3328

1

VIEWS

DOWNLOADS

CITATIONS

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Source: Figshare

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USAGE METRICS

2341	3328	1
VIEWS	DOWNLOADS	CITATIONS 

[Learn more about usage metrics](#) 



Source: Figshare

Named Entity Recognition (NER)

MD simulations of P-glycoprotein MOL started from three different crystal structures: 3G5U MOL , 4M1M MOL and 4KSB MOL . Molecular simulations of mouse P-glycoprotein MOL started from three different crystal structures with PDB IDs corresponding to 3G5U MOL , 4KSB MOL and 4M1M MOL . The protein was transferred to POPC MOL / cholesterol MOL bilayer, followed by energy minimization and 10 ns equilibration period. All the simulations were performed in Gromacs SOFTNAME 3.3.3 SOFTVERS in conjunction with GROMOS54a7 FFM force field at 300 K TEMP and pressure of 1 bar. Each system was run in triplicates for 200 ns STIME .

Strategies based on language models:

- Encoder-like models (BERT): ~ 100 millions parameters (SLM)
→ *fine tuning* (many manually annotated examples)
- Decoder-like models (GPT): ~ 10-1000 billions parameters (LLM)
→ ***harness*** (instructions + constraints) / **access to models?**

Manual approach

OpenAI, GPT-5.5, 10/06/2026

In the following text, extract all molecule names, software versions, software names, forcefield or model names, simulation temperatures, and simulation times (duration):

MD simulations of P-glycoprotein started from three different crystal structures: 3G5U, 4M1M and 4KSB

Molecular simulations of mouse P-glycoprotein started from three different crystal structures with PDB IDs corresponding to 3G5U, 4KSB and 4M1M. The protein was transferred to POPC/cholesterol bilayer, followed by energy minimization and 10 ns equilibration period. All the simulations were performed in Gromacs 3.3.3 in conjunction with GROMOS54a7 force field at 300 K and pressure of 1 bar. Each system was run in triplicates for 200 ns.

Afficher moins ^

Manual approach

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Afficher moins ^

Extracted entities from the text:

Category	Extracted Value
Molecule names	P-glycoprotein (mouse P-glycoprotein), POPC, cholesterol
Software names	Gromacs
Software versions	Gromacs 3.3.3
Forcefield / model names	GROMOS54a7 force field
Simulation temperatures	300 K
Simulation times (duration)	10 ns equilibration period; 200 ns production simulation (each system, run in triplicate)

Additional identifiers present (not requested but mentioned):

- Crystal structure / PDB IDs: 3G5U, 4KSB, 4M1M
- Pressure: 1 bar

Manual approach

OpenAI, GPT-5.5, 10/06/2026

- Automation?
- Model selection?
- Detailed instructions?
- Output format?
- Quality control?

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Automation & model selection: OpenRouter

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[Get API Key](#)[Explore Models ↗](#)

100T

Monthly Tokens

8M+

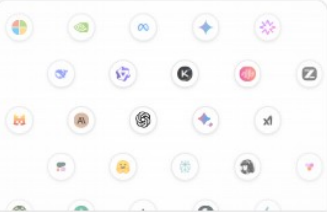
Global Users

60+

Providers

400+

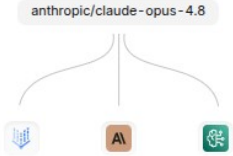
Models



One API for Any Model

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Custom Data Policies

Protect your organization with fine grained data policies. Ensure prompts only go to the models and providers you trust.

[View docs ↗](#)

<https://openrouter.ai/>

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Automation & model selection: OpenRouter

AI

Anthropic: Claude Sonnet 4.6

anthropic/claude-sonnet-4.6

Sonnet 4.6 is Anthropic's most capable Sonnet-class model, designed for professional work. It excels at iterative development, code generation, and complex reasoning tasks.

Show more

MODALITIES

T I A → T

IN / OUT PRICE

\$3 / \$15

Qwen: Qwen3.6 27B

qwen/qwen3.6-27b

Qwen3.6 27B is a dense 27-billion-parameter language model from the Qwen Team at Alibaba, released in April 2026. It features hybrid multimodal capabilities — accepting text, image, and video inputs — and supports a 262,144-token context window.

Show more

MODALITIES

T I A → T

IN / OUT PRICE

\$0,289 / \$2,40 per 1M

Avg

CONTEXT

262K

High

RELEASED

Apr 27, 2026

Google: Gemini 3.1 Pro Preview

google/gemini-3.1-pro-preview

Gemini 3.1 Pro Preview is Google's frontier reasoning model, designed for improved performance on complex tasks and code generation.

Show more

MODALITIES

T I A → T

IN / OUT PRICE

\$0,12 / \$0,35 per 1M

Low

CONTEXT

262K

High

RELEASED

Apr 2, 2026

Google: Gemma 4 31B

google/gemma-4-31b-1t

Gemma 4 31B Instruct is Google DeepMind's 30.7B parameter model, designed for improved performance on complex tasks and code generation.

Show more

MODALITIES

T I A → T

IN / OUT PRICE

\$0,435 / \$0,87 per 1M

Low

CONTEXT

1M

High

RELEASED

Jun 9, 2026

Anthropic: Claude Fable 5

anthropic/claude-fable-5

Claude Fable 5 is a Mythos-class model from Anthropic, designed for improved performance on complex tasks and code generation.

Show more

MODALITIES

T I A → T

IN / OUT PRICE

\$10 / \$50 per 1M

High

CONTEXT

1M

High

RELEASED

Jun 9, 2026

DeepSeek: DeepSeek V4 Pro

deepseek/deepseek-v4-pro

DeepSeek V4 Pro is a large-scale Mixture-of-Experts model from DeepSeek with 1.6T total parameters and 490B activated parameters, supporting a 1M-token context window. It is designed for advanced reasoning, coding, and complex tasks.

Show more

MODALITIES

T → T

IN / OUT PRICE

\$0,435 / \$0,87 per 1M

Low

CONTEXT

1M

High

RELEASED

Jun 9, 2026

P. Poulain | CC BY

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API

```
import requests
import json

# First API call with reasoning
response = requests.post(
    url="https://openrouter.ai/api/v1/chat/completions",
    headers={
        "Authorization": "Bearer <OPENROUTER_API_KEY>",
        "Content-Type": "application/json",
    },
    data=json.dumps({
        "model": "qwen/qwen3.7-plus",
        "messages": [
            {
                "role": "user",
                "content": "How many r's are in the word 'strawberry'?"
            }
        ],
        "reasoning": {"enabled": True}
    })
)

# Extract the assistant message with reasoning_details
response = response.json()
response = response['choices'][0]['message']
```

Prompt engineering

In the following text, extract all molecule names, software versions, software names, forcefield or model names, simulation temperatures, and simulation times (duration):

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Afficher moins ^

Prompt engineering

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Afficher moins ^

Text to annotate

~ 3000 tokens

E. Touami_{16 / 31}

16 / 31

Generic instructions

```
# Named-Entity Recognition task

## Role definition

You are a highly specialized "Named Entity Recognition (NER) engine" dedicated to
extracting entities from Molecular Dynamics (MD) simulation dataset descriptions or
research articles.

Your sole function is to analyze the input text and output a valid JSON object containing
only the extracted entities and their categories.
```

```
## Annotation rules

### Molecule: MDL

#### Definition

The "MDL" entity covers molecular compounds, including simple molecules, ions, nucleic acids, proteins, lipids, sugars, polymers, and complexes.
MDL entities should be uniquely identifiable.
```

```

see Rules
- Exclude whitespaces and punctuation marks around entities
- Annotate both singular and plural forms
- Include chemical formulas and abbreviations
- Include amino acid and nucleic acid sequences
- Include any identifiers (from 1b to 1d, 1f, 1g, 1h, 1i, 1j, 1k, 1l, 1m, 1n, 1o, 1p, 1q, 1r, 1s, 1t, 1u, 1v, 1w, 1x, 1y, 1z, 1aa, 1ab, 1ac, 1ad, 1ae, 1af, 1ag, 1ah, 1ai, 1aj, 1ak, 1al, 1am, 1an, 1ao, 1ap, 1aq, 1ar, 1as, 1at, 1au, 1av, 1aw, 1ax, 1ay, 1az, 1ba, 1bb, 1bc, 1bd, 1be, 1bf, 1bg, 1bh, 1bi, 1bj, 1bk, 1bl, 1bm, 1bn, 1bo, 1bp, 1bq, 1br, 1bs, 1bt, 1bu, 1bv, 1bw, 1bx, 1by, 1bz, 1ca, 1cb, 1cc, 1cd, 1ce, 1cf, 1cg, 1ch, 1ci, 1cj, 1ck, 1cl, 1cm, 1cn, 1co, 1cp, 1cq, 1cr, 1cs, 1ct, 1cu, 1cv, 1cw, 1cx, 1cy, 1cz, 1da, 1db, 1dc, 1dd, 1de, 1df, 1dg, 1dh, 1di, 1dj, 1dk, 1dl, 1dm, 1dn, 1do, 1dp, 1dq, 1dr, 1ds, 1dt, 1du, 1dv, 1dw, 1dx, 1dy, 1dz, 1ea, 1eb, 1ec, 1ed, 1ee, 1ef, 1eg, 1eh, 1ei, 1ej, 1ek, 1el, 1em, 1en, 1eo, 1ep, 1eq, 1er, 1es, 1et, 1eu, 1ev, 1ew, 1ex, 1ey, 1ez, 1fa, 1fb, 1fc, 1fd, 1fe, 1ff, 1fg, 1fh, 1fi, 1fj, 1fk, 1fl, 1fm, 1fn, 1fo, 1fp, 1fq, 1fr, 1fs, 1ft, 1fu, 1fv, 1fw, 1fx, 1fy, 1fz, 1ga, 1gb, 1gc, 1gd, 1ge, 1gf, 1gg, 1gh, 1gi, 1gj, 1gk, 1gl, 1gm, 1gn, 1go, 1gp, 1gq, 1gr, 1gs, 1gt, 1gu, 1gv, 1gw, 1gx, 1gy, 1gz, 1ha, 1hb, 1hc, 1hd, 1he, 1hf, 1hg, 1hh, 1hi, 1hj, 1hk, 1hl, 1hm, 1hn, 1ho, 1hp, 1hq, 1hr, 1hs, 1ht, 1hu, 1hv, 1hw, 1hx, 1hy, 1hz, 1ia, 1ib, 1ic, 1id, 1ie, 1if, 1ig, 1ih, 1ii, 1ij, 1ik, 1il, 1im, 1in, 1io, 1ip, 1iq, 1ir, 1is, 1it, 1iu, 1iv, 1iw, 1ix, 1iy, 1iz, 1ja, 1jb, 1jc, 1jd, 1je, 1jf, 1jg, 1jh, 1ji, 1jj, 1jk, 1jl, 1jm, 1jn, 1jo, 1jp, 1jq, 1jr, 1js, 1jt, 1ju, 1jv, 1jw, 1jx, 1jy, 1jz, 1ka, 1kb, 1kc, 1kd, 1ke, 1kf, 1kg, 1kh, 1ki, 1kj, 1kk, 1kl, 1km, 1kn, 1ko, 1kp, 1kq, 1kr, 1ks, 1kt, 1ku, 1kv, 1kw, 1kx, 1ky, 1kz, 1la, 1lb, 1lc, 1ld, 1le, 1lf, 1lg, 1lh, 1li, 1lj, 1lk, 1ll, 1lm, 1ln, 1lo, 1lp, 1lq, 1lr, 1ls, 1lt, 1lu, 1lv, 1lw, 1lx, 1ly, 1lz, 1ma, 1mb, 1mc, 1md, 1me, 1mf, 1mg, 1mh, 1mi, 1mj, 1mk, 1ml, 1mm, 1mn, 1mo, 1mp, 1mq, 1mr, 1ms, 1mt, 1mu, 1mv, 1mw, 1mx, 1my, 1mz, 1na, 1nb, 1nc, 1nd, 1ne, 1nf, 1ng, 1nh, 1ni, 1nj, 1nk, 1nl, 1nm, 1nn, 1no, 1np, 1nq, 1nr, 1ns, 1nt, 1nu, 1nv, 1nw, 1nx, 1ny, 1nz, 1oa, 1ob, 1oc, 1od, 1oe, 1of, 1og, 1oh, 1oi, 1oj, 1ok, 1ol, 1om, 1on, 1oo, 1op, 1oq, 1or, 1os, 1ot, 1ou, 1ov, 1ow, 1ox, 1oy, 1oz, 1pa, 1pb, 1pc, 1pd, 1pe, 1pf, 1pg, 1ph, 1pi, 1pj, 1pk, 1pl, 1pm, 1pn, 1po, 1pp, 1pq, 1pr, 1ps, 1pt, 1pu, 1pv, 1pw, 1px, 1py, 1pz, 1qa, 1qb, 1qc, 1qd, 1qe, 1qf, 1qg, 1qh, 1qi, 1qj, 1qk, 1ql, 1qm, 1qn, 1qo, 1qp, 1qq, 1qr, 1qs, 1qt, 1qu, 1qv, 1qw, 1qx, 1qy, 1qz, 1ra, 1rb, 1rc, 1rd, 1re, 1rf, 1rg, 1rh, 1ri, 1rj, 1rk, 1rl, 1rm, 1rn, 1ro, 1rp, 1rq, 1rr, 1rs, 1rt, 1ru, 1rv, 1rw, 1rx, 1ry, 1rz, 1sa, 1sb, 1sc, 1sd, 1se, 1sf, 1sg, 1sh, 1si, 1sj, 1sk, 1sl, 1sm, 1sn, 1so, 1sp, 1sq, 1sr, 1ss, 1st, 1su, 1sv, 1sw, 1sx, 1sy, 1sz, 1ta, 1tb, 1tc, 1td, 1te, 1tf, 1tg, 1th, 1ti, 1tj, 1tk, 1tl, 1tm, 1tn, 1to, 1tp, 1tq, 1tr, 1ts, 1tt, 1tu, 1tv, 1tw, 1tx, 1ty, 1tz, 1ua, 1ub, 1uc, 1ud, 1ue, 1uf, 1ug, 1uh, 1ui, 1uj, 1uk, 1ul, 1um, 1un, 1uo, 1up, 1uq, 1ur, 1us, 1ut, 1uu, 1uv, 1uw, 1ux, 1uy, 1uz, 1va, 1vb, 1vc, 1vd, 1ve, 1vf, 1vg, 1vh, 1vi, 1vj, 1vk, 1vl, 1vm, 1vn, 1vo, 1vp, 1vq, 1vr, 1vs, 1vt, 1vu, 1vv, 1vw, 1vx, 1vy, 1vz, 1wa, 1wb, 1wc, 1wd, 1we, 1wf, 1wg, 1wh, 1wi, 1wj, 1wk, 1wl, 1wm, 1wn, 1wo, 1wp, 1wq, 1wr, 1ws, 1wt, 1wu, 1wv, 1ww, 1wx, 1wy, 1wz, 1xa, 1xb, 1xc, 1xd, 1xe, 1xf, 1xg, 1xh, 1xi, 1xj, 1xk, 1xl, 1xm, 1xn, 1xo, 1xp, 1xq, 1xr, 1xs, 1xt, 1xu, 1xv, 1xw, 1xx, 1xy, 1xz, 1ya, 1yb, 1yc, 1yd, 1ye, 1yf, 1yg, 1yh, 1yi, 1yj, 1yk, 1yl, 1ym, 1yn, 1yo, 1yp, 1yq, 1yr, 1ys, 1yt, 1yu, 1yv, 1yw, 1yx, 1yy, 1yz, 1za, 1zb, 1zc, 1zd, 1ze, 1zf, 1zg, 1zh, 1zi, 1zj, 1zk, 1zl, 1zm, 1zn, 1zo, 1zp, 1zq, 1zr, 1zs, 1zt, 1zu, 1zv, 1zw, 1zx, 1zy, 1zz, 1aa, 1ab, 1ac, 1ad, 1ae, 1af, 1ag, 1ah, 1ai, 1aj, 1ak, 1al, 1am, 1an, 1ao, 1ap, 1aq, 1ar, 1as, 1at, 1au, 1av, 1aw, 1ax, 1ay, 1az, 1ba, 1bb, 1bc, 1bd, 1be, 1bf, 1bg, 1bh, 1bi, 1bj, 1bk, 1bl, 1bm, 1bn, 1bo, 1bp, 1bq, 1br, 1bs, 1bt, 1bu, 1bv, 1bw, 1bx, 1by, 1bz, 1ca, 1cb, 1cc, 1cd, 1ce, 1cf, 1cg, 1ch, 1ci, 1cj, 1ck, 1cl, 1cm, 1cn, 1co, 1cp, 1cq, 1cr, 1cs, 1ct, 1cu, 1cv, 1cw, 1cx, 1cy, 1cz, 1da, 1db, 1dc, 1dd, 1de, 1df, 1dg, 1dh, 1di, 1dj, 1dk, 1dl, 1dm, 1dn, 1do, 1dp, 1dq, 1dr, 1ds, 1dt, 1du, 1dv, 1dw, 1dx, 1dy, 1dz, 1ea, 1eb, 1ec, 1ed, 1ee, 1ef, 1eg, 1eh, 1ei, 1ej, 1ek, 1el, 1em, 1en, 1eo, 1ep, 1eq, 1er, 1es, 1et, 1eu, 1ev, 1ew, 1ex, 1ey, 1ez, 1fa, 1fb, 1fc, 1fd, 1fe, 1ff, 1fg, 1fh, 1fi, 1fj, 1fk, 1fl, 1fm, 1fn, 1fo, 1fp, 1fq, 1fr, 1fs, 1ft, 1fu, 1fv, 1fw, 1fx, 1fy, 1fz, 1ga, 1gb, 1gc, 1gd, 1ge, 1gf, 1gg, 1gh, 1gi, 1gj, 1gk, 1gl, 1gm, 1gn, 1go, 1gp, 1gq, 1gr, 1gs, 1gt, 1gu, 1gv, 1gw, 1gx, 1gy, 1gz, 1ha, 1hb, 1hc, 1hd, 1he, 1hf, 1hg, 1hh, 1hi, 1hj, 1hk, 1hl, 1hm, 1hn, 1ho, 1hp, 1hq, 1hr, 1hs, 1ht, 1hu, 1hv, 1hw, 1hx, 1hy, 1hz, 1ia, 1ib, 1ic, 1id, 1ie, 1if, 1ig, 1ih, 1ii, 1ij, 1ik, 1il, 1im, 1in, 1io, 1ip, 1iq, 1ir, 1is, 1it, 1iu, 1iv, 1iw, 1ix, 1iy, 1iz, 1ja, 1jb, 1jc, 1jd, 1je, 1jf, 1jg, 1jh, 1ji, 1jj, 1jk, 1jl, 1jm, 1jn, 1jo, 1jp, 1jq, 1jr, 1js, 1jt, 1ju, 1jv, 1jw, 1jx, 1jy, 1jz, 1ka, 1kb, 1kc, 1kd, 1ke, 1kf, 1kg, 1kh, 1ki, 1kj, 1kk, 1kl, 1km, 1kn, 1ko, 1kp, 1kq, 1kr, 1ks, 1kt, 1ku, 1kv, 1kw, 1kx, 1ky, 1kz, 1la, 1lb, 1lc, 1ld, 1le, 1lf, 1lg, 1lh, 1li, 1lj, 1lk, 1ll
```

```
#### Good examples
+ 'sodium chloride'
+ 'ethanol'
+ 'ammonia'
+ 'Q29537'
+ 'ZRN1'
```

```

- Na+
#### Bad examples
- 'lipids'
- 'DNA'
- 'hydrated sodium chloride' (only annotate 'sodium chloride')
- 'binding protein'
- 'ions in solvent'

```

```
### Force field and model: FFM

#### Definition

The "FFM" entity refers to any force field or molecular model used to describe the interactions between particles in a simulation. This includes all classical all-atom force fields, coarse-grained models, solvent models, and water models. Both the name and version of the force field/model are relevant and should be annotated when available.
```

```
#### Rules

. Include water models and other specific solvent models (e.g., 'TIP3P', 'SPC/E')
. Exclude generic terms like 'force field' or 'model'

#### Good examples

. 'CHARMM36'
. 'AMBER99SB'
```

Annotation guide

```
- 'GROMOS5368'  
- 'GROMOS94 43A1'  
- 'TIP3P'  
  
#### Bad examples  
  
- 'the force field'  
- 'TIP3P water' (only annotate 'TIP3P')  
- 'the CHARMM36 force field' (only annotate 'CHARMM36')
```

```
## Software name: SOFTNAME

### Definition

The "SOFTNAME" entity refers to the name of any software used for molecular simulation, visualization, or analysis. It includes packages for molecular dynamics, modeling, trajectory processing, and any other computational tasks relevant to the simulation workflow.
```

```

#see Rules

- Exclude generic words such as 'software', 'tool', or 'program' unless they are part of
  the official name.
- Exclude algorithms like 'SHAKE', 'RESP'...

### Good examples

- 'GROMACS'
- VMD
- NAMD
- PYMOL
- siRNA

```

```

- 'COLVAR'
### Bad examples
- 'Python'
- 'the simulation software'
- 'GROMACS software' (only annotate 'GROMACS')

```

```
new Software version SOFTWARES

#### Definition

The "SOFTWARES" entity refers to the version identifier of any software used in the
simulation process.
It includes version numbers, release tags, or labels, regardless of formatting (e.g.,
numeric, date-based, semantic).

#### Rules
```

- Must follow a corresponding `"SOFTWARE"` entity (software/tool name)
- Numeric and symbolic parts intact (e.g., '1.2.1-beta')
- Include any suffixes (e.g., '2020.4'), prefixes (e.g., 'v5.1.2')

Good examples

- 'v5.8'
- '2020.4'
- '5.1.4'

```
### Bad examples
- 'latest version'
- 'software (v. 2018.4)' (only annotate 'v. 2018.4')
- 'release 2023.1' (only annotate '2023.1')

### Simulation time: STIME

### Definition
```

Examples (few-shot prompting)

Output format

```
The "STIME" entity refers to the duration for which a production molecular dynamics simulation is run.
```

```
### Rules
```

```
- Exclude minimization or equilibration time
```

```
10-50% of the entire expression if it refers to time
```

The unit is not mandatory, but the context must unambiguously indicate that the number refers to a simulation time

Good examples

- 5×200
- 50 picoseconds
- 100 ns
- 4.5 μ s
- 10 to 50 ns

```

### Bad example
- 'for several hours of computation'
- 10-20 repls

### Simulation temperature: STMP

### Definition

The "STMP" property refers to the thermal conditions under which a simulation is
conducted. It includes explicitly stated temperature values, with or without units.

```

```
#### Rules

- The unit is not mandatory, but the context must unambiguously indicate that the number refers to temperature
- Exclude surrounding words like 'temperature of' or 'heated to'
- Include 'has temperature' or 'body temperature'
```

```
### Good examples
- '300K'
- '500 degrees Celsius'
- '298'
- 'room temperature'

### Bad examples
- 'heater up'
```

```
## Output Format (Mandatory)
ABSOLUTE: DO NOT output any text, explanation, reasoning, or markdown (e.g., ```json).
The output "## MUST" be a valid JSON object where the list of entities is contained within
the key "entities".

Expected output structure:
```

```
{
  "entities": [
    {
      "category": "MOL",
      "text": "cholesterol",
    },
    {
      "category": "FFM",
      "text": "MARTINI2",
    },
    {
      "category": "MOL",
      "text": "POPC",
    },
    {
      "category": "MOL",
      "text": "glycerol",
    },
    {
      "category": "SOFTWARE",
      "text": "Gromacs",
    }
  ]
}
```

```
## Trajectories (few-shot demonstrations)
## Example 1
Input text:
Short molecular dynamics of a peptide inside a pure DMPC membrane\n\n ms of molecular
dynamics simulation of a 10-residue peptide inside a pure DMPC membrane, performed at 300
K using ensemble -I at a 1 nm using Berendsen (semi-isotropic) and V-rescale for pressure
and temperature coupling, respectively, with 1.0 ps and 0.10 ps coupling constants. Variables
```

per Waals interactions were treated using the Verlet algorithm with a 1 nm cutoff. Electrostatics were treated with particle-mesh Ewald, with a 1 nm cutoff for the real space calculations. Both the peptide and DMPc were parameterized using the GROMOS 54a7, while SPC was used for water. LINCS restraints were used on all the bonds. Peptide sequence is AAAQAQAQAQAQAQAQAQAQA. This sequence was not taken from any real world examples, as this is a test system created for MDAnalysis datasets. The peptide was set to have a α -helical secondary structure, and was inserted manually in the membrane before minimization and equilibration.

```
{
  "entities": [
    {
      "category": "MOL",
      "text": "DMPC"
    },
    {
      "category": "STEM",
      "text": "1 ms"
    },
    {
      "category": "MOL",
      "text": "DMPC"
    },
    {
      "category": "STEM",
      "text": "216K"
    },
    {
      "category": "MOL",
      "text": "DMPC"
    },
    {
      "category": "FFM",
      "text": "GROMOS 54A1"
    },
    {
      "category": "FFM",
      "text": "SCC"
    },
    {
      "category": "MOL",
      "text": "AQUADAGMAHQAGATPQA"
    },
    {
      "category": "SOFTWARE",
      "text": "MDAnalysis"
    }
  ]
}
```

```
### Example 2

Input text:
Unbiased simulation of Alanine Dipeptide in gas phase\n57 microsecond long unbiased
Molecular Dynamics (MD) trajectory of alanine dipeptide in gas phase [traj comp.xtc].
Temperature 300 K. Force Field: AMBER99SB-ILDN. There are 30+ back and forth
transitions between the C 7eq and the C 7ax state. Simulations were performed using
```

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```
Output:
{"entities":
  [
    {"category": "MOL", "text": "Alanine dipeptide"},
    {"category": "TIME", "text": "67 microsecond"},
    {"category": "MOL", "text": "alanine dipeptide"},
    {"category": "STEM", "text": "354 x3"},
    {"category": "PFW", "text": "ANR80255-110M"},
    {"category": "SOFTWARE", "text": "GROMACS"}
  ]
}
```

```
{
  "category": "SOPV1825", "text": "2021.1",
  "category": "SOPV1825", "text": "COLVAV",
  "category": "MOL", "text": "Alanine dipeptide"
}

## Example 3
Input text:
```

GROMOS 43A1-53 POPE simulations (versions 1 and 2) 313 K (NOTE: anisotropic pressure coupling) (note: GROMOS 43A1-53 POPE bilayer simulations performed using gromops 4.0.7.4 to 40 ns with different starting velocities) Simulations were performed with the standard GROMOS 43A1-53 settings: a 1.0 fs cut-off with PME for the Coulombic interactions and a twin-range 1.0/1.6 fs cut-off for the van der Waals interactions. These simulations were performed at 313 K with a 128 lipid bilayer and used anisotropic pressure coupling. The trajectories are provided for the initial 160 ns. The starting structure was made up through the conversion of an equilibrated GROMOS 43A1-53 POPE membrane.

```

    "category": "FEW", "text": "GRONOS 43A1-53"},
    "category": "MOL", "text": "POPE"},
    "category": "STEMP", "text": "313 K"},
    "category": "FEW", "text": "GRONOS 43A1-53"},
    "category": "MOL", "text": "POPE"},
    "category": "SOFTWARE", "text": "GRONMACS"},
    "category": "SOFTWARE", "text": "4.0.7"}
  ]
}
```

```
{
  "category": "STIME", "text": "200 ns",
  "category": "STIME", "text": "213 ns",
  "category": "STIME", "text": "180 ns"},
  "category": "FEF", "text": "GRMONG 43A1-51"},
  "category": "mol", "text": "POPC"}
}
```

Task to Execute

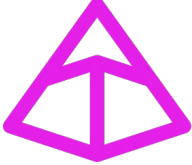
Identify and Extract Molecular Dynamics simulation entities in the following text:

MD simulations of *P. galeatus* started from three different crystal structures: 3G5U

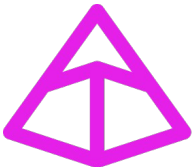
Molecular simulations of mouse P-glycoprotein started from three different crystal structures with PDB IDs corresponding to 365U, 4XSB and 4M1M. The protein was transferred to POPC/cholesterol bilayer, followed by energy minimization and 10 ns equilibration period. All the simulations were performed in Gromacs 3.3.3 in conjunction with GROMOSS407 force field at 300 K and pressure of 1 bar. Each system was run in triplicate for 200 ns.

Automation, output format & quality control

Automation, output format & quality control

Data model  **Pydantic**

Automation, output format & quality control

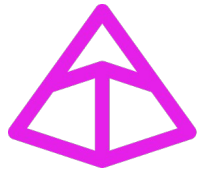
Data model  **Pydantic**

```
class Entity(BaseModel):
    category: str = Field(..., description="Category identifying the entity type.")
    text: str = Field(..., description="Extracted text content.")

class Molecule(Entity):
    category: Literal["MOL"] = Field(
        "MOL", description="Category for molecule entities."
    )
```

Automation, output format & quality control

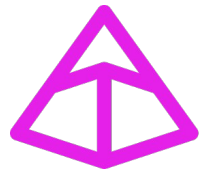
Data model



Pydantic

Automation, output format & quality control

Data model



Pydantic

Forces LLM to output
the data model



Instructor

Results (best of)

Model name	Release date	Parameters (B)	Precision	Unitary cost (\$)	Unitary inf. time (s)
anthropic/claude-sonnet-4.6	2026/02	?	0.89	0.002	0.5
z-ai/glm-5.1	2026/04	754	0.88	0.001	5.6
qwen/qwen3.6-27b	2026/04	27	0.87	0.002	8.0
google/gemini-3.1-pro-preview	2026/02	?	0.85	0.004	2.5
minimax/minimax-m2.7	2026/03	229	0.85	> 0.001	3.1
google/gemma-4-31b-it	2026/04	31	0.84	> 0.001	1.1
openai/gpt-5.5	2026/04	?	0.84	0.003	1.5

Results (worst of)

Model name	Release date	Parameters (B)	Precision	Unitary cost (\$)	Unitary inf. time (s)
deepseek/deepseek-chat	2024/12	685	0.74	< 0.001	2.2
mistralai/mistral-small-3.2-24b-instruct	2025/06	24	0.73	< 0.001	0.2
mistralai/mixtral-8x22b-instruct	2024/04	141	0.68	0.001	0.2
mistralai/ministral-8b-2512	2025/12	8	0.67	< 0.001	0.4
meta-llama/llama-3.1-8b-instruct	2024/07	8	0.56	< 0.001	1.0

Simulation time normalization

MD simulations of **P-glycoprotein MOL** started from three different crystal structures: **3G5U MOL** , **4M1M MOL** and **4KSB MOL** . Molecular simulations of mouse **P-glycoprotein MOL** started from three different crystal structures with PDB IDs corresponding to **3G5U MOL** , **4KSB MOL** and **4M1M MOL** . The protein was transferred to **POPC MOL** / **cholesterol MOL** bilayer, followed by energy minimization and 10 ns equilibration period. All the simulations were performed in **Gromacs SOFTNAME** **3.3.3 SOFTVERS** in conjunction with **GROMOS54a7 FFM** force field at **300 K TEMP** and pressure of 1 bar. Each system was run in triplicates for **200 ns STIME** .

Simulation time normalization

MD simulations of **P-glycoprotein MOL** started from three different crystal structures: **3G5U MOL** , **4M1M MOL** and **4KSB MOL** . Molecular simulations of mouse **P-glycoprotein MOL** started from three different crystal structures with PDB IDs corresponding to **3G5U MOL** , **4KSB MOL** and **4M1M MOL** . The protein was transferred to **POPC MOL** / **cholesterol MOL** bilayer, followed by energy minimization and 10 ns equilibration period. All the simulations were performed in **Gromacs SOFTNAME** **3.3.3 SOFTVERS** in conjunction with **GROMOS54a7 FFM** force field at **300 K TEMP** and pressure of 1 bar. Each system was run in triplicates for **200 ns STIME** .

- 200ns
- 350 ns
- 50 ns
- 6 μ s

Simulation time normalization

MD simulations of **P-glycoprotein MOL** started from three different crystal structures: **3G5U MOL** , **4M1M MOL** and **4KSB MOL** . Molecular simulations of mouse **P-glycoprotein MOL** started from three different crystal structures with PDB IDs corresponding to **3G5U MOL** , **4KSB MOL** and **4M1M MOL** . The protein was transferred to **POPC MOL** / **cholesterol MOL** bilayer, followed by energy minimization and 10 ns equilibration period. All the simulations were performed in **Gromacs SOFTNAME** **3.3.3 SOFTVERS** in conjunction with **GROMOS54a7 FFM** force field at **300 K TEMP** and pressure of 1 bar. Each system was run in triplicates for **200 ns STIME** .

- 200ns
- 350 ns
- 50 ns
- 6 μ s

Regular expressions (Regex)

([0-9]+) **(\.[0-9]+)?** ***(ps|ns| μ s|ms|s)**

Simulation time normalization

MD simulations of **P-glycoprotein MOL** started from three different crystal structures: **3G5U MOL** , **4M1M MOL** and **4KSB MOL** . Molecular simulations of mouse **P-glycoprotein MOL** started from three different crystal structures with PDB IDs corresponding to **3G5U MOL** , **4KSB MOL** and **4M1M MOL** . The protein was transferred to **POPC MOL** / **cholesterol MOL** bilayer, followed by energy minimization and 10 ns equilibration period. All the simulations were performed in **Gromacs SOFTNAME** **3.3.3 SOFTVERS** in conjunction with **GROMOS54a7 FFM** force field at **300 K TEMP** and pressure of 1 bar. Each system was run in triplicates for **200 ns STIME** .

- 200ns
- 350 ns
- 50 ns
- 6μs

Regular expressions (Regex)

([0-9]+) **(\\.?[0-9]+)?** ***(ps|ns|μs|ms|s)**



- 5-microsecond
- 200-300ns
- one hundred nanosecond
- 1.5 micro-sec

Simulation time normalization



MD simulations of P-glycoprotein MOL started from three different crystal structures: 3G5U MOL , 4M1M MOL and 4KSB MOL . Molecular simulations of mouse P-glycoprotein MOL started from three different crystal structures with PDB IDs corresponding to 3G5U MOL , 4KSB MOL and 4M1M MOL . The protein was transferred to POPC MOL / cholesterol MOL bilayer, followed by energy minimization and 10 ns equilibration period. All the simulations were performed in Gromacs SOFTNAME 3.3.3 SOFTVERS in conjunction with GROMOS54a7 FFM force field at 300 K TEMP and pressure of 1 bar. Each system was run in triplicates for 200 ns STIME .

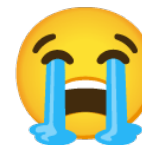
- 200ns
- 350 ns
- 50 ns
- 6μs

Regular expressions (Regex)

`([0-9]+)(\\.?[0-9]+)?*(ps|ns|μs|ms|s)`



- 5-microsecond
- 200-300ns
- one hundred nanosecond
- 1.5 micro-sec



Simulation time normalization

Model: DeepSeek V4 Pro

Prompt:

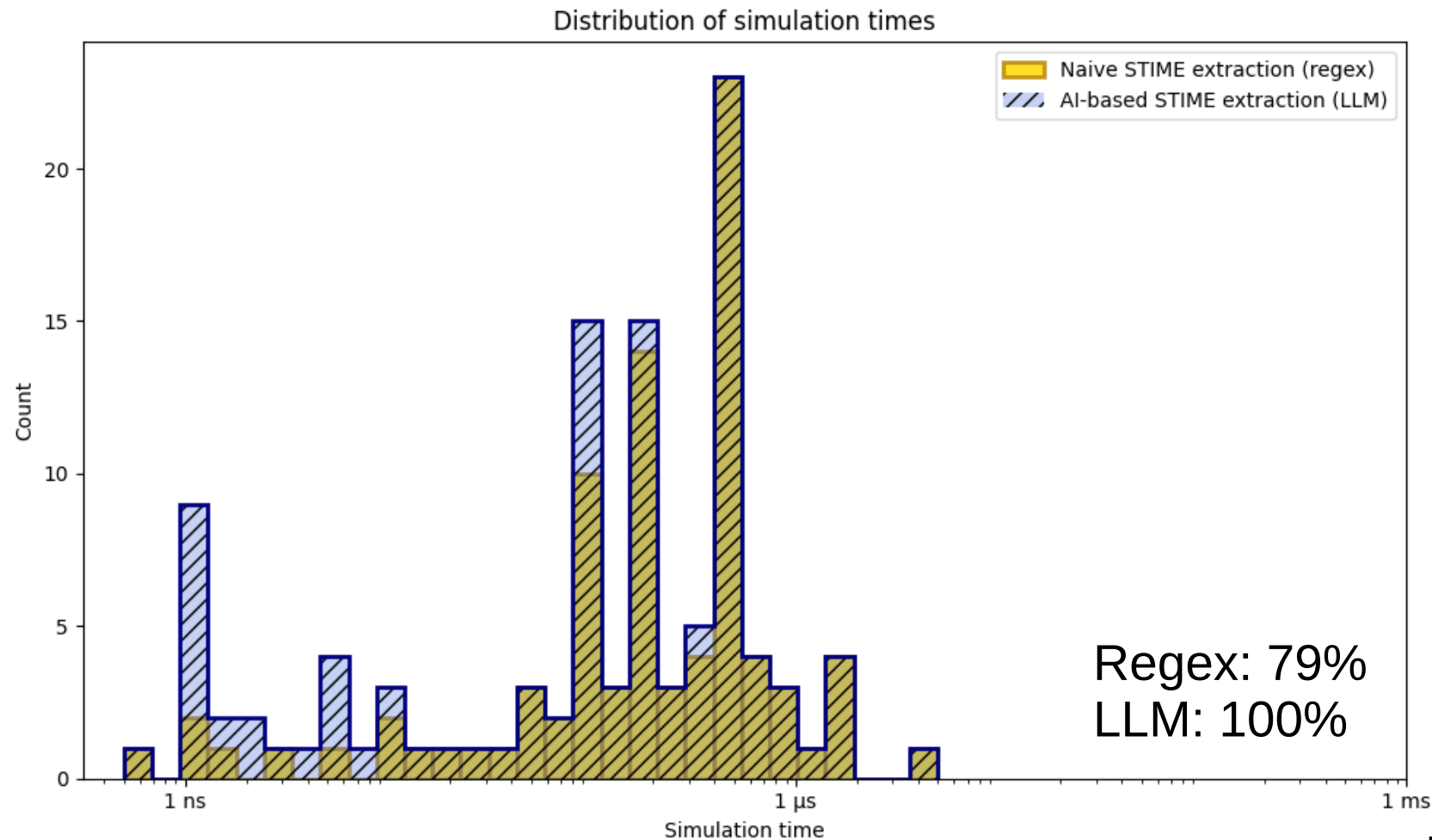
You are a unit normalization assistant for molecular dynamics simulation times.
Your tasks:

- Convert all time units to standard time abbreviations (ps, ns, μ s, ms, s)
- Separate numerical values from time units

Rules:

- No markdown, no explanation
- Use only standard time units: ps (picoseconds), ns (nanoseconds), μ s (microseconds), ms (milliseconds), s (seconds)
- Always separate value and unit (e.g. "500ns" $\rightarrow$ value: 500, unit: "ns")
- Take in consideration values written in letter (e.g. "one hundred"), and convert it to numeric value
- If the simulation time is an interval, separate each simulation time in the interval.
- If the unit is missing or the unit is not a time unit, output the normalized unit to "None"
- If the numerical value is missing, output the normalized value to "None"

Simulation time normalization



Conclusion

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Metadata extraction and normalization are key for the reuse of open research data (and publications).

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Large language models (e.g., ChatGPT) can be used as research tools, provided they are **equipped with sufficient harness**.

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Large language models (e.g., ChatGPT) can be used as research tools, provided they are **equipped with sufficient harness**.

The field of LLMs is evolving rapidly:

- Most small models released last year or two-years ago perform significantly worse than current SOTA models.
- Access to large, up-to-date models (**proprietary and open-weight**) is important.

Discussion

Access to large, up-to-date models (proprietary and open-weight)

What I don't want to do:

- Read benchmarks (mostly biased and not corresponding to my use case)
- Find the “best” model
- Find how much VRAM (GPU) is needed to run the model (whichllm)
- Find how to run the model (vLLM)
- Connect to a computer cluster:
 - Wait to get access to GPU
 - Download an (open-weight) model
 - Wait...
 - Run inferences
- Repeat

Local ‘computer’

```
(base) pierre@banana 🇧🇪 ~ $ uvx whichllm@latest plan "deepseek v4 pro"
```

Model Info

Model: deepseek-ai/DeepSeek-V4-Pro
Params: 1600.0B (49.0B active) | Arch: deepseek | Context: unknown
License: mit

VRAM Required (context: 4096)

Quant	VRAM	Quality Loss
Q2_K	473.0 GB	-25%
Q3_K_M	659.2 GB	-8%
Q4_K_M ★	845.5 GB	-5%
Q5_K_M	1031.8 GB	-3%
Q6_K	1218.0 GB	-2%
Q8_0	1590.6 GB	-1%
F16	2987.5 GB	0%

Local ‘computer’

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(base) pierre@banana 📶 ~ $ uvx whichllm@latest plan "deepseek v4 pro"
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Q8_0	1590.6 GB	-1%
F16	2987.5 GB	0%

LDLC

GARANTIE 3 ANS INCLUSE

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Chercher un produit, une marque, une caté

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6

VOTRE PANIER :

DÉSIGNATION	PRIX	QUANTITÉ	SOUS-TOTAL	
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TOTAL

179 999€⁷⁵ HT

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PASSER COMMANDE

API Hugging Face: open-weight only

🔥 Inference Providers · Metrics for top trending models

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📘 Learn more

Filter by model or provider...

Model	Provider	Input \$/1M	Output \$/1M	Context	Latency(s)	Throughput(t/s)	Tools	Structured
nvidia/NVIDIA-Nemotron-3-Ultra-550B-A55...	together	\$0.60	\$3.60	512,288	0.31	106	Yes	No
deepseek-ai/DeepSeek-V4-Pro	novita	\$1.60	\$3.38	1,048,576	0.40	37	Yes	No
deepseek-ai/DeepSeek-V4-Pro	together	\$1.74	\$3.48	512,000	0.60	50	Yes	Yes
deepseek-ai/DeepSeek-V4-Pro	fireworks-ai	-	-	1,048,576	1.05	66	Yes	No
deepseek-ai/DeepSeek-V4-Pro	deepinfra	\$1.74	\$3.48	65,536	1.43	29	Yes	Yes
Qwen/Qwen3.6-35B-A3B	deepinfra	\$0.15	\$0.95	262,144	0.33	131	Yes	Yes
google/gemma-4-31B-it	novita	\$0.14	\$0.40	262,144	0.65	79	Yes	Yes
google/gemma-4-31B-it	together	\$0.39	\$0.97	262,144	0.29	41	Yes	Yes
google/gemma-4-31B-it	deepinfra	\$0.13	\$0.38	262,144	0.39	21	Yes	Yes
deepseek-ai/DeepSeek-V4-Flash	novita	\$0.14	\$0.28	1,048,576	0.75	98	Yes	No
deepseek-ai/DeepSeek-V4-Flash	fireworks-ai	-	-	-	1.00	78	Yes	No
deepseek-ai/DeepSeek-V4-Flash	deepinfra	\$0.14	\$0.28	1,048,576	1.09	25	Yes	Yes
Qwen/Qwen3.6-27B	ovhcloud	-	-	-	0.36	73	Yes	Yes
google/gemma-4-26B-A4B-it	novita	\$0.13	\$0.40	262,144	0.66	31	No	Yes
google/gemma-4-26B-A4B-it	deepinfra	\$0.07	\$0.34	262,144	0.50	39	Yes	Yes
moonshotai/Kimi-K2.6	novita	\$0.80	\$3.40	262,144	1.17	25	Yes	No
moonshotai/Kimi-K2.6	together	\$1.20	\$4.50	262,144	0.39	51	Yes	Yes
moonshotai/Kimi-K2.6	fireworks-ai	-	-	262,144	0.35	99	No	No
moonshotai/Kimi-K2.6	deepinfra	\$0.75	\$3.50	262,144	0.64	18	Yes	No

<https://huggingface.co/inference/models>

API Albert

Des modèles pour tous vos usages

Albert API offre l'accès à une large gamme de modèle fondation open source d'IA générative. **Nous mettons à jour régulièrement les modèles disponibles pour vous permettre d'utiliser les modèles qui font l'état de l'art.**

D'où proviennent les modèles disponibles ?

Ces modèles ne sont pas propres à l'administration, il s'agit de modèle mis à disposition par des tiers comme Mistral ou Meta sur la plateforme 🤖 [HuggingFace](#) 🔗. Hébergé sur nos serveurs, aucune de vos données n'est envoyée à ces fournisseurs de modèles.

120B, 08/2025



APACHE 2.0

OPENWEIGHT-LARGE

openai/gpt-oss-120b

Modèle de chat pour des tâches complexes





24B, 06/2025



APACHE 2.0

OPENWEIGHT-MEDIUM

mistralai/Mistral-Small-3.2-24B-Instruct-2506

Modèle de chat pour des tâches d'une complexité modérée et d'analyse d'images





8B, 12/2025



APACHE 2.0

OPENWEIGHT-SMALL

mistralai/Mistral-3-8B-Instruct-2512

Modèle de chat pour des tâches simples





<https://ia.numerique.gouv.fr/outils-ia/albert-api/>

API Scaleway



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All requests performed using [Batches API](#) are priced with a -50% discount.

Select Region



Paris

Name ↕	Tasks ↕	Input tokens ↕	Output tokens ↕	
qwen3.5-397b-a17b	Chat and code	€0.60 / MILLION TOKENS	€3.60 / MILLION TOKENS	Try
qwen3.6-35b-a3b	Chat and Vision	€0.25 / MILLION TOKENS	€1.50 / MILLION TOKENS	Try
gemma-4-26b-a4b-it	Chat and Vision	€0.25 / MILLION TOKENS	€0.50 / MILLION TOKENS	Try
gpt-oss-120b	Chat	€0.15 / MILLION TOKENS	€0.60 / MILLION TOKENS	Try
mistral-medium-3.5-128b	Chat and Vision	€1.50 / MILLION TOKENS	€7.50 / MILLION TOKENS	Try
whisper-large-v3	Audio transcription	€0.003 / AUDIO MINUTE	Free	Try
llama-3.3-70b-instruct	Chat	€0.90 / MILLION TOKENS	€0.90 / MILLION TOKENS	Try
qwen3-235b-a22b-instruct-2507	Chat	€0.75 / MILLION TOKENS	€2.25 / MILLION TOKENS	Try

<https://www.scaleway.com/en/pricing/model-as-a-service/>

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